

Exploratory Data Analysis

4/17/26

Adam Finnimore, Jacob Sellitti

Contents

```
library(tidyverse)
library(reticulate)
reticulate::py_install("pyarrow")
```

Using virtual environment '/Users/jacob/.virtualenvs/r-reticulate' ...

```
import pandas as pd
import matplotlib.pyplot as plt
```

Exploring the Data

What is your outcome variable(s)? How well does it measure the outcome you are interested? How does it relate to your expectations? What are your key explanatory variables?

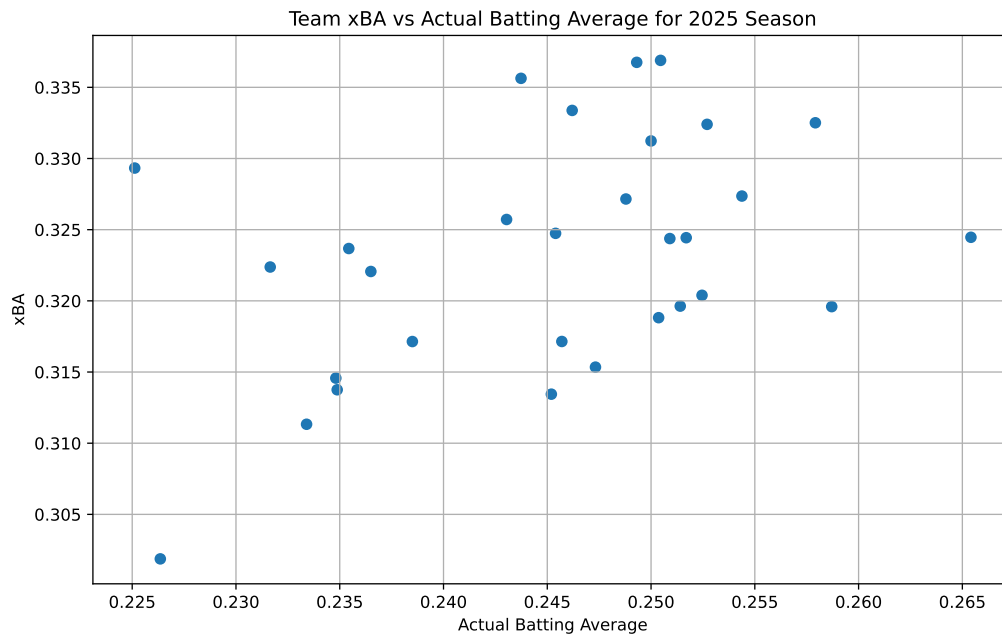
Our outcome variable for this project is xBA (Expected Batting Average). This metric is used to calculate the expected batting average of a batted ball, and subsequently the likelihood that it results in a hit. For instance, a home run well beyond the fence would have an xBA of 1.000, since it will result in a hit 100% of the time. xBA has been shown to be a useful metric, and when used in conjunction with actual batting average (AVG), provides an additional layer of complexity in quantifying hitting performance. This relationship can be visualized in the graphs below, plotting AVG vs xBA to determine which players and teams have been the luckiest or unluckiest.

```

all_pitches = pd.read_parquet("../data/2025_all_pitches.parquet")
data = pd.read_parquet("../data/2025_hits.parquet")
team_data = {}
for team in data['team'].unique():
    single_data = data[data['team'] == team]
    team_xba = single_data['xBA'].mean()
    team_strikeouts_count = len(all_pitches[
        ((all_pitches['inning_topbot'] == 'Top') & (all_pitches['away_team'] == team) |
         (all_pitches['inning_topbot'] == 'Bot') & (all_pitches['home_team'] == team)) &
        (all_pitches['events'] == 'strikeout')
    ])
    team_avg = len(single_data[single_data['hit'] == 1]) / (len(single_data) + team_strikeouts_count)
    team_data[team] = [team_xba, team_avg]

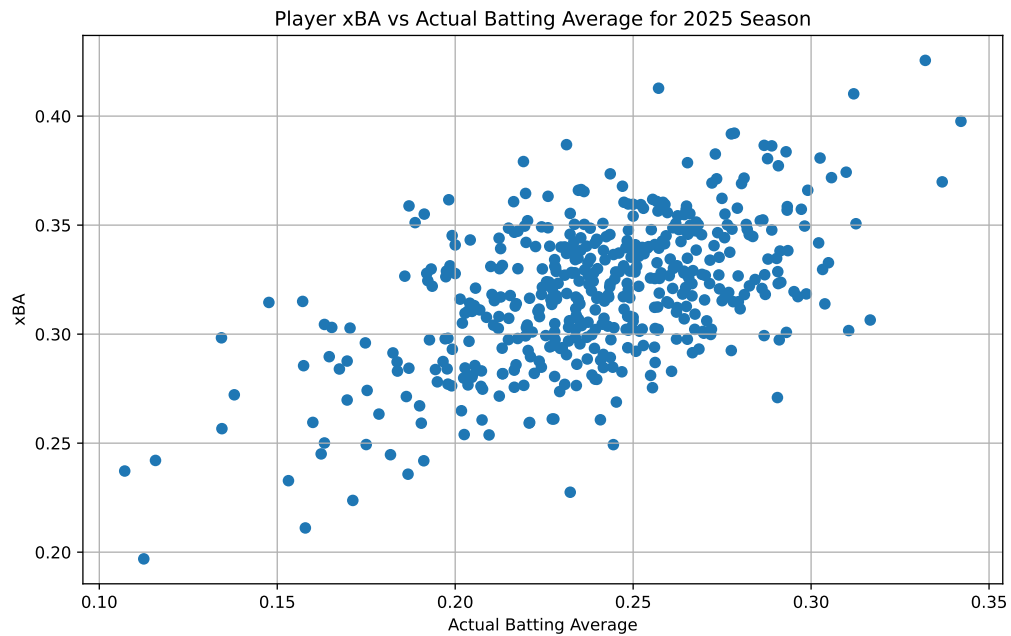
team_df = pd.DataFrame.from_dict(team_data, orient='index', columns=['xBA', 'avg'])
team_df = team_df.sort_values(by='xBA', ascending=False)
plt.figure(figsize=(10, 6))
plt.scatter(team_df['avg'], team_df['xBA'])
plt.xlabel('Actual Batting Average')
plt.ylabel('xBA')
plt.title('Team xBA vs Actual Batting Average for 2025 Season')
plt.grid()
plt.show()

```

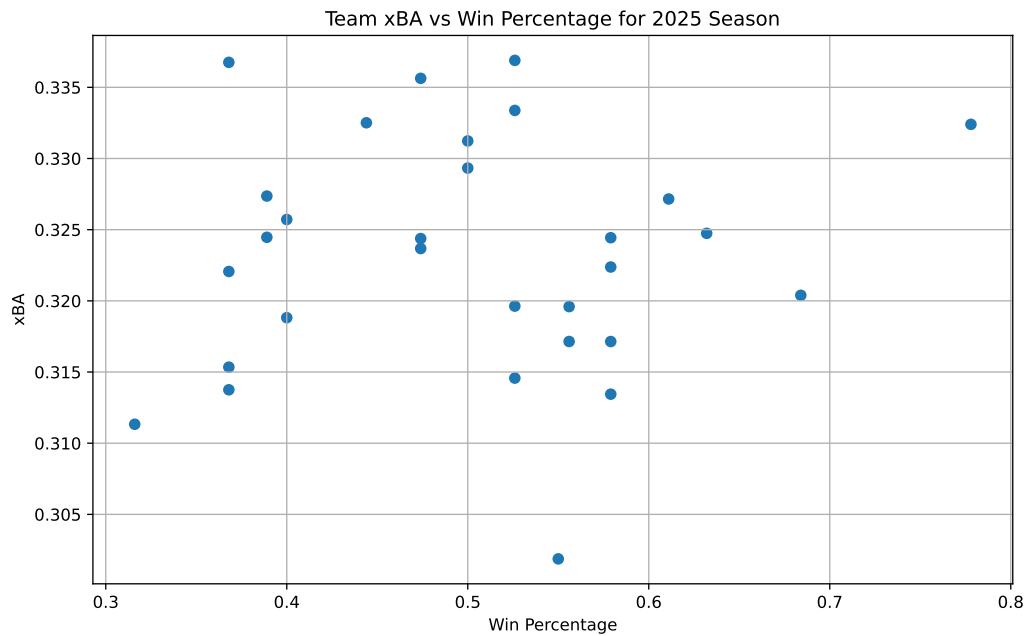


```
batter_data = {}
# Only use batters with more than 50 plate appearances
batter_list = data['batter'].value_counts()
batter_list = batter_list[batter_list > 50].index.tolist()
for batter in batter_list:
    single_data = data[data['batter'] == batter]
    batter_xba = single_data['xBA'].mean()
    batter_strikeouts_count = len(all_pitches[(all_pitches['batter'] == batter) & (all_pitches['strikeouts'] == 1)])
    batter_avg = len(single_data[single_data['hit'] == 1]) / (len(single_data) + batter_strikeouts_count)
    batter_data[batter] = [batter_xba, batter_avg]

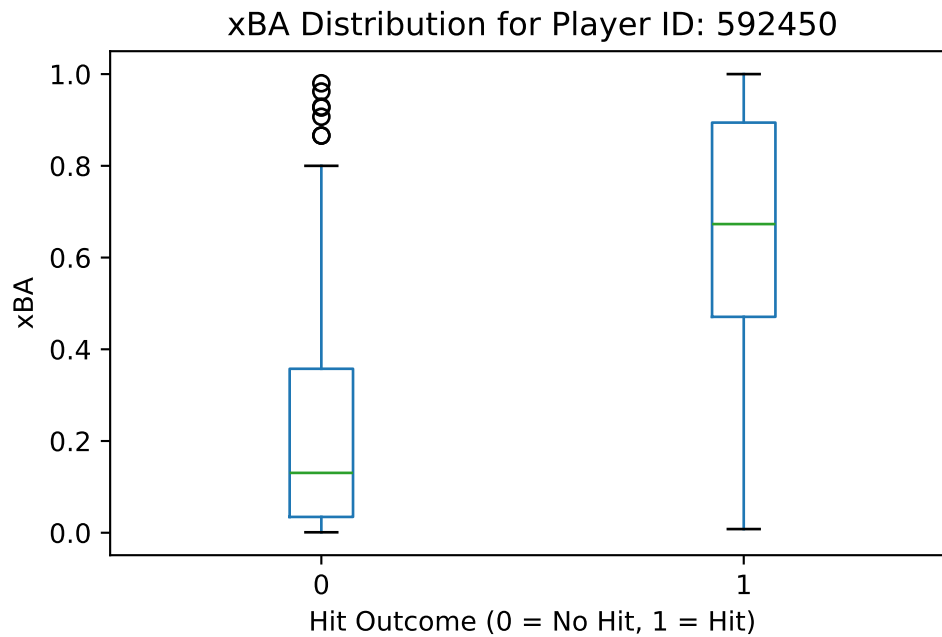
batter_df = pd.DataFrame.from_dict(batter_data, orient='index', columns=['xBA', 'avg'])
batter_df = batter_df.sort_values(by='xBA', ascending=False)
plt.figure(figsize=(10, 6))
plt.scatter(batter_df['avg'], batter_df['xBA'])
plt.xlabel('Actual Batting Average')
plt.ylabel('xBA')
plt.title('Player xBA vs Actual Batting Average for 2025 Season')
plt.grid()
plt.show()
```



```
team_stats = pd.read_csv("../data/2025_team_stats.csv")
team_df['Win%'] = team_stats.set_index('Team').loc[team_df.index]['Win%']
plt.figure(figsize=(10, 6))
plt.scatter(team_df['Win%'], team_df['xBA'])
plt.xlabel('Win Percentage')
plt.ylabel('xBA')
plt.title('Team xBA vs Win Percentage for 2025 Season')
plt.grid()
plt.show()
```



```
player_id = 592450 # Aaron Judge
player_data = data[data['batter'] == player_id]
plt.figure(figsize=(8, 6))
player_data.boxplot(column='xBA', by='hit')
plt.title(f'xBA Distribution for Player ID: {player_id}')
plt.suptitle('')
plt.xlabel('Hit Outcome (0 = No Hit, 1 = Hit)')
plt.ylabel('xBA')
plt.grid()
plt.show()
```



```
plt.close()
```

The key explanatory variables in calculating xBA are exit velocity and launch angle. This high-level metrics can be immediately useful in predicting how far a batted ball will travel, and in what direction. With the MLB's advanced Statcast systems, the calculation also includes park-specific information such as field dimensions, and the batter's sprint speed. However, for the purposes of this project, we will also be utilizing data such as pitch speed, rotation, and the location where the ball landed. These are all theoretically useful features for predicting the hit probability, but could introduce unnecessary complexity and data inconsistency (since hit location coordinate systems are not centered at the same place in every ballpark).

Create a table of summary statistics for the variables you are planning to use.

```
data = pd.read_parquet("../data/2025_hits.parquet")
df = data[['launch_speed', 'launch_angle', 'hit_distance',
          'hit_location_x', 'hit_location_y',
          'pitch_speed', 'release_spin_rate', 'xBA']]
summary = df.describe()
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 80)
print(summary)
```

	launch_speed	launch_angle	hit_distance	hit_location_x	\
count	122071.0	122071.0	122071.0	122071.0	
mean	88.997012	13.131464	165.640406	126.624398	
std	14.720204	28.627369	134.919862	40.564939	
min	5.6	-89.0	0.0	2.05	
25%	80.5	-5.0	21.0	101.52	
50%	91.9	13.0	159.0	125.48	
75%	100.1	31.0	287.0	153.85	
max	120.4	90.0	485.0	246.29	

	hit_location_y	pitch_speed	release_spin_rate	xBA
count	122071.0	122071.0	122071.0	122071.0
mean	122.584466	89.611042	2255.709005	0.323245
std	41.101798	6.401187	364.398434	0.286012
min	4.75	28.1	42.0	0.001
25%	90.56	85.6	2115.0	0.067
50%	123.81	90.6	2294.0	0.249
75%	156.59	94.4	2464.0	0.522
max	242.89	105.4	3501.0	1.0

Data Wrangling and Transformation:

What data cleaning did you have to do? How did you wrangle the data? Are you deciding to exclude any observations? If so, why? Did you have to create any new variables from existing variables? If so, how and why?

The data we pulled was already very clean and was not missing any important variables for any observations. The data contains a list of every pitch that was thrown throughout the season, but since we are analyzing batting averages and efficiency, we removed all observations that did not result in a ball being put into play, as these do not have the impact metrics that we are looking to analyze. The data came with a game id variable that we split up to obtain the home team, away team, and stadium from. We wanted to take the stadium, as different ballparks have different field lengths and fence heights, so this will have an effect on many of our hits. By obtaining the home and away teams, we are able to look at full team data, as well as possibly compare performances at home vs. on the road for teams.